Deep Reinforcement Learning Based Resource Allocation in a Simplified 5G Downlink Network

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1. Motivation & Problem Statement

The exponential growth of mobile data traffic in 5G networks demands intelligent and adaptive resource management strategies. Traditional schedulers such as Round Robin and Proportional Fair rely on fixed heuristics and fail to adapt to the dynamic nature of modern wireless channels. This project addresses the Radio Resource Management (RRM) problem in 5G downlink networks by applying Deep Reinforcement Learning (DRL) to optimize the allocation of Physical Resource Blocks (PRBs) among multiple User Equipments (UEs), jointly maximizing system throughput and fairness.

2. Objectives

- Design and implement a simplified 5G downlink simulation with a realistic channel model.
- Formulate resource allocation as a Markov Decision Process (MDP) and train a Deep Q-Network (DQN) agent.
- Maximize system throughput while maintaining user fairness via Jain's Fairness Index.
- Benchmark DQN against Round Robin, Proportional Fair, and Random Allocation baselines.

3. Technical Approach

Network Model	Single-cell downlink: 1 gNodeB, $N = 10$–20 UEs, 50–100 Resource Blocks (RBs).
Channel Model	Log-distance path loss: $\mathrm{PL}(d) = \mathrm{PL}_0 + 10n\log_{10}(d)$. Per-user SNR computed; achievable rate via Shannon's formula $C = B\log_2(1+\mathrm{SNR})$.
State Space	Per-user SNR, current throughput, and RB occupancy vector. Total dimension ≈ 20–40.
Action Space	RB assignment per UE at each scheduling interval, with ε-greedy exploration policy.
Reward Function	$R = \alpha \sum_i C_i \; + \; \beta \cdot J, \quad \text{where } J = \dfrac{(\sum_i x_i)^2}{n \sum_i x_i^2}$ is Jain's Fairness Index.
DQN Architecture	PyTorch fully-connected network with ReLU activations, Experience Replay buffer, and periodic Target Network updates for stable training.

4. System Architecture

5. Project Timeline (8 Weeks)

Week 1–2	5G environment setup: channel model, SINR computation, RB pool simulation.
Week 3–4	Baseline schedulers: Round Robin, Proportional Fair, Random Allocation.
Week 5–6	DQN agent: neural network, experience replay buffer, target network (PyTorch).
Week 7–8	Training, evaluation, comparison plots (throughput, fairness, convergence), final report.

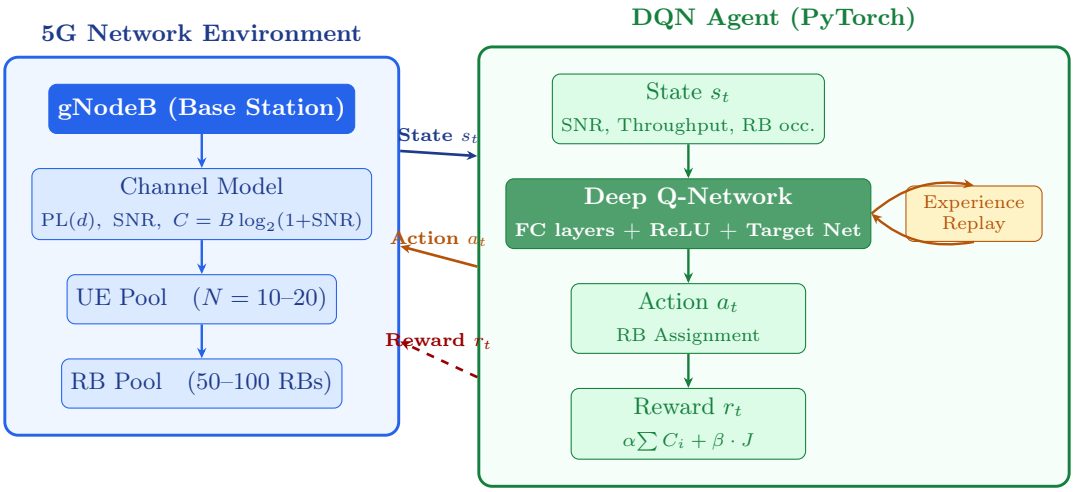


Figure 1: System architecture of the proposed DRL-based 5G downlink resource allocation framework. The environment exposes state observations to the DQN agent, which selects RB assignments and receives a scalar reward combining throughput and fairness.

6. Tools & Implementation

Language: Python 3.10+ | **RL Framework:** PyTorch | **Simulation:** NumPy, Matplotlib | **Team:** 2 members | **Duration:** 8 weeks